

Consequently, there is a constant $C \in \mathbb{R}_+$ such that for a block A “above the diagonal” and with corner points on the grid Π_n we have

$$\mathbb{E}[|\sqrt{n}U_n(h_2)(A)|^2] \leq C \frac{|A|}{n} \leq C|A|^{3/2}, \quad (3.40)$$

where the last step follows because $\sqrt{|A|} = \sqrt{(u-s)(v-t)} \geq \sqrt{n^{-1}n^{-1}} = n^{-1}$.

(b) *A block “on the diagonal”.* In this case $s = t < u = v$ and therefore (cf. (3.35))

$$\begin{aligned} & n^2(n-1)^2 \mathbb{E}[|U_n(h_2)(A)|^2] \\ &= \sum_{i,k,j,\ell=\lfloor ns \rfloor+1}^{\lfloor nu \rfloor} |\mathbb{E}[h_2(X_i, X_j)h_2(X_k, X_\ell)]|. \end{aligned} \quad (3.41)$$

Once more, taking into account the symmetry of the problem, we only need to consider the cases where $i \leq k$. Furthermore, it is sufficient to consider only the following three instances

$$(i) \quad i < j \leq k < \ell, \quad (ii) \quad i \leq k \leq j \leq \ell, \quad (iii) \quad i \leq k < \ell \leq j.$$

For example, the first case also encompasses the cases $j < i \leq k < \ell$ and $i < j \leq \ell < k$ and $j < i \leq \ell < k$ because

$$\begin{aligned} \mathbb{E}[h_2(X_i, X_j)h_2(X_k, X_\ell)] &= \mathbb{E}[h_2(X_j, X_i)h_2(X_k, X_\ell)] \\ &= \mathbb{E}[h_2(X_i, X_j)h_2(X_\ell, X_k)] = \mathbb{E}[h_2(X_j, X_i)h_2(X_\ell, X_k)]. \end{aligned}$$

Now the cases (i), (ii) and (iii) essentially work in the same spirit and again rely heavily on the property of h_2 being degenerate as well as on the knowledge of the relative positions of the indices i, j, k, ℓ .

Case (i): Using the stationarity of the process, we obtain

$$\begin{aligned} & \sum_{\lfloor ns \rfloor+1 \leq i < j \leq k < \ell \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_i, X_j)h_2(X_k, X_\ell)]| \\ &= \sum_{\lfloor ns \rfloor+1 \leq i < j \leq k < \ell \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_{j-i})h_2(X_{k-i}, X_{\ell-i})]| \\ &\leq n(u-s) \sum_{\lfloor ns \rfloor+1 \leq o, r, w \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})]|. \end{aligned} \quad (3.42)$$

We split the sum in (3.42) in three subsums depending on the relative positions of the three indices: (α) $o, r \leq w$, (β) $r, w \leq o$ and (γ) $o, w \leq r$. Depending on the subcase, we proceed similar as in the setting for a block “above the diagonal” and use a suitable coupling in combination with (3.34) to derive a fast enough decay of the expectation.

Subcase (α) :

$$\begin{aligned}
& \sum_{\lfloor ns \rfloor + 1 \leq o, r \leq w \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})]| \\
&= \sum_{\lfloor ns \rfloor + 1 \leq o, r \leq w \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})] - \mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w}^{(w)})]| \\
&\lesssim \sum_{\lfloor ns \rfloor + 1 \leq o, r \leq w \leq \lfloor nu \rfloor} \mathbb{E}[d(X_0, X_o^{(w)})^{\rho p}]^{1/p} \\
&\lesssim n(u-s) \sum_{w=1}^{\infty} w \mathbb{E}[d(X_0, X_o^{(w)})^{\rho p}]^{1/p}.
\end{aligned}$$

Subcase (β) :

$$\begin{aligned}
& \sum_{\lfloor ns \rfloor + 1 \leq r, w \leq o \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})]| \\
&\leq \sum_{\lfloor ns \rfloor + 1 \leq r \leq w \leq o \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})]| \\
&\quad + \sum_{\lfloor ns \rfloor + 1 \leq w \leq r \leq o \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})]| \\
&\leq \sum_{\lfloor ns \rfloor + 1 \leq r \leq w \leq o \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})] \\
&\quad - \mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w}^{(w)})]| \\
&\quad + \sum_{\lfloor ns \rfloor + 1 \leq w \leq r \leq o \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})] \\
&\quad - \mathbb{E}[h_2(X_0, X_o^{(o)})h_2(X_{o+r}, X_{o+r+w}^{(r)})]| \\
&\lesssim n(u-s) \sum_{w=1}^{\infty} w \mathbb{E}[d(X_0, X_o^{(w)})^{\rho p}]^{1/p} + n(u-s) \sum_{o=1}^{\infty} o \mathbb{E}[d(X_0, X_o^{(o)})^{\rho p}]^{1/p} \\
&\quad + n(u-s) \sum_{r=1}^{\infty} r \mathbb{E}[d(X_0, X_o^{(r)})^{\rho p}]^{1/p}.
\end{aligned}$$

Subcase (γ) :

$$\begin{aligned}
& \sum_{\lfloor ns \rfloor + 1 \leq o, w \leq r \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})]| \\
&\leq \sum_{\lfloor ns \rfloor + 1 \leq o \leq w \leq r \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})] \\
&\quad - \mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w}^{(w)})]| \\
&+ \sum_{1 \leq w \leq v \leq u \leq n} |\mathbb{E}[h_2(X_0, X_o)h_2(X_{o+r}, X_{o+r+w})]|
\end{aligned}$$

$$\begin{aligned}
& - \mathbb{E}[h_2(X_0, X_0^{(o)})h_2(X_{o+r}^{(r)}, X_{o+r+w}^{(r)})] \\
& \lesssim n(u-s) \sum_{w=1}^{\infty} w \mathbb{E}[d(X_0, X_0^{(w)})^{\rho p}]^{1/p} + n(u-s) \sum_{o=1}^{\infty} o \mathbb{E}[d(X_0, X_0^{(o)})^{\rho p}]^{1/p} \\
& \quad + n(u-s) \sum_{r=1}^{\infty} r \mathbb{E}[d(X_0, X_0^{(r)})^{\rho p}]^{1/p}.
\end{aligned}$$

Thus, collecting the three subcases and returning to (3.42), we see once more with the approximating property from Equation (2.7) that

$$\sum_{\lfloor ns \rfloor + 1 \leq i < j \leq k < \ell \leq \lfloor nu \rfloor} |\mathbb{E}[h_2(X_i, X_j)h_2(X_k, X_\ell)]| \lesssim n^2(u-s)^2.$$

One finds with similar ideas that in

Case (ii):

$$\sum_{1 \leq i \leq k \leq j \leq \ell \leq n} |\mathbb{E}[h_2(X_i, X_j)h_2(X_k, X_\ell)]| \lesssim n^2(u-s)^2;$$

Case (iii):

$$\sum_{1 \leq i \leq k < \ell \leq j \leq n} |\mathbb{E}[h_2(X_i, X_j)h_2(X_k, X_\ell)]| \lesssim n^2(u-s)^2.$$

Consequently, there is a constant $C \in \mathbb{R}_+$ such that for a block A “on the diagonal” and with corner points on the grid Γ_n we have

$$\mathbb{E}[|\sqrt{n}U_n(h_2)(A)|^2] \leq C \frac{|A|}{n} \leq C|A|^{3/2}. \quad (3.43)$$

This completes the proof. $\square$

Proof of Corollary 2.6. Observe that the 2-dimensional increment of $U_n(h_2)$ over $A = (0, 1] \times (0, 1]$ is simply

$$U_n(h_2)(A) = U_n(h_2)(1, 1) = U_n(h_2) = \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} h_2(X_i, X_j).$$

Using the upper bound (3.43) which is derived in the proof Theorem 2.5 we conclude

$$\begin{aligned}
\mathbb{E} \left[|\sqrt{n} U_n(h_2)(A)|^2 \right] &= \mathbb{E} \left[\left| \frac{2}{\sqrt{n}(n-1)} \sum_{1 \leq i < j \leq n} h_2(X_i, X_j) \right|^2 \right] \\
&= \frac{4}{n(n-1)^2} \sum_{1 \leq i < j \leq n} \sum_{1 \leq k < \ell \leq n} \mathbb{E}[h_2(X_i, X_j)h_2(X_k, X_\ell)] \leq \frac{C}{n}
\end{aligned}$$

for a certain $C \in \mathbb{R}_+$. $\square$